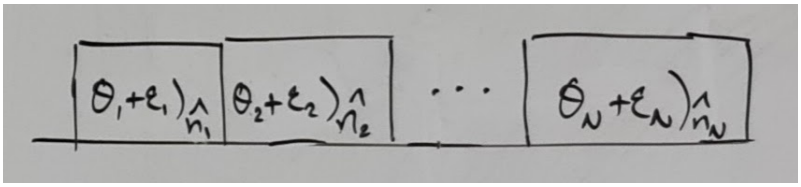
Rotation Errors in Composite Pulse Sequences

Sahand Tabatabaei

1 Conventions and Setup

- We use the Pauli vector $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. The unit vector $\hat{\mathbf{n}}_j$ specifies the axis of pulse j. Matrix products are denoted by $\prod_{j=1}^{N} B_j = B_N \cdots B_1$ and $\overleftarrow{\prod}_{j=1}^{N} B_j = B_1 \cdots B_N$.
- Consider a sequence of δ-pulses with rotation angles $\theta_j + \epsilon_j$, where $\epsilon_j \ll 1$ is a rotation error. We want to find a perturbative expansion for the propagator in powers of $\{\epsilon_j\}_{j=1}^{N}$.



$$U(\boldsymbol{\epsilon}) = \overleftarrow{\prod_{j=1}^{N}} e^{-i(\theta_j+\epsilon_j)\hat{\mathbf{n}}_j\cdot\boldsymbol{\sigma}/2} = \overleftarrow{\prod_{j=1}^{N}} \left(e^{-i\theta_j\hat{\mathbf{n}}_j\cdot\boldsymbol{\sigma}/2} e^{-i\epsilon_j\hat{\mathbf{n}}_j\cdot\boldsymbol{\sigma}/2}\right). \tag{1}$$

2 Separating the Ideal Operation From the Errors

- Define the shorthands $P_j = e^{-i\theta_j\hat{\mathbf{n}}_j\cdot\boldsymbol{\sigma}/2}$, $E_j = e^{-i\epsilon_j\hat{\mathbf{n}}_j\cdot\boldsymbol{\sigma}/2}$ and $U_0 = P_N \cdots P_1$. Factor out the unperturbed unitary by inserting resolutions of the identity:

$$\begin{aligned} U(\boldsymbol{\epsilon}) &= P_N E_N \cdots P_2 E_2 P_1 E_1 \\ &= \underbrace{P_N \cdots P_1}_{U_0} \big(P_1^\dagger \cdots P_{N-1}^\dagger E_N P_{N-1} \cdots P_1\big) \times \big(P_1^\dagger \cdots P_{N-2}^\dagger E_{N-1} P_{N-2} \cdots P_1\big) \times \cdots \times \big(P_1^\dagger P_2^\dagger E_3 P_2 P_1\big)\big(P_1^\dagger E_2 P_1\big) E_1. \end{aligned}$$

- Apply the similarity transformations on E_j. Let $R_{\hat{\mathbf{n}}}(\theta) \in \mathrm{SO}(3)$ denote the rotation matrix corresponding to $e^{-i\theta\hat{\mathbf{n}}\cdot\boldsymbol{\sigma}/2}$. Then

$$\boxed{U(\boldsymbol{\epsilon}) = U_0 \overleftarrow{\prod_{j=1}^{N}} e^{-i\epsilon_j\hat{\mathbf{m}}_j\cdot\boldsymbol{\sigma}/2}}, \tag{2}$$

where $U_0 = \prod_{j=1}^{N} e^{-i\theta_j\hat{\mathbf{n}}_j\cdot\boldsymbol{\sigma}/2}$ is the unperturbed total unitary and $\hat{\mathbf{m}}_j \equiv \left(\prod_{k=1}^{j-1} R_{\hat{\mathbf{n}}_k}(\theta_k)\right)^T \hat{\mathbf{n}}_j$, where $\hat{\mathbf{m}}_1 \equiv \hat{\mathbf{n}}_1$ is understood for $j = 1$. For the first few pulses, this is

$$\begin{aligned} \hat{\mathbf{m}}_1 &= \hat{\mathbf{n}}_1, \\ \hat{\mathbf{m}}_2 &= R_{\hat{\mathbf{n}}_1}^T(\theta_1)\hat{\mathbf{n}}_2, \\ \hat{\mathbf{m}}_3 &= \left[R_{\hat{\mathbf{n}}_2}(\theta_2) R_{\hat{\mathbf{n}}_1}(\theta_1)\right]^T \hat{\mathbf{n}}_3. \end{aligned}$$

So the unitary is a product of the unperturbed unitary U_0 and individual rotations by angles ϵ_j around the axes $\hat{\mathbf{m}}_j$. Each $\hat{\mathbf{m}}_j$ is obtained by rotating $\hat{\mathbf{n}}_j$ by the inverse sequence of unperturbed rotations.

3 Magnus Expansion of the Error Rotation

- Now that we have isolated a product of exponentials with small exponents, we can apply a Magnus expansion to approximate them as a single exponential $U(\boldsymbol{\epsilon}) = U_0 e^{-i\boldsymbol{\beta}(\boldsymbol{\epsilon})\cdot\boldsymbol{\sigma}/2}$. Write

$$\prod_{j=1}^{N} e^{-i\epsilon_j \hat{\mathbf{m}}_j \cdot \boldsymbol{\sigma}/2} = e^{A(\boldsymbol{\epsilon})}.$$

- To order $\mathcal{O}(\epsilon^2)$, the Magnus expansion gives

$$A(\boldsymbol{\epsilon}) = -i\sum_{j=1}^{N} \epsilon_j \frac{\hat{\mathbf{m}}_j \cdot \boldsymbol{\sigma}}{2} + \frac{(-i)^2}{2}\sum_{k=2}^{N}\sum_{j=1}^{k-1} \frac{\epsilon_j \epsilon_k}{4} \underbrace{[\hat{\mathbf{m}}_k \cdot \boldsymbol{\sigma}, \hat{\mathbf{m}}_j \cdot \boldsymbol{\sigma}]}_{2i(\hat{\mathbf{m}}_k \times \hat{\mathbf{m}}_j)\cdot\boldsymbol{\sigma}} + \mathcal{O}(\epsilon^3), \tag{3}$$

  and so we identify the error rotation vector

$$\boxed{\boldsymbol{\beta}(\boldsymbol{\epsilon}) = \sum_{j=1}^{N} \epsilon_j \hat{\mathbf{m}}_j + \frac{1}{2}\sum_{k=2}^{N}\sum_{j=1}^{k-1} \epsilon_k \epsilon_j (\hat{\mathbf{m}}_k \times \hat{\mathbf{m}}_j) + \mathcal{O}(\epsilon^3).} \tag{4}$$

4 Application to Robust Composite Sequences

- *Robustness of the unitary:*

$$\sum_{j=1}^{N} \epsilon_j \hat{\mathbf{m}}_j = \mathbf{0} \qquad \text{(first order)}, \tag{5}$$

$$\sum_{k=2}^{N}\sum_{j=1}^{k-1} \epsilon_k \epsilon_j \hat{\mathbf{m}}_k \times \hat{\mathbf{m}}_j = \mathbf{0} \qquad \text{(second order)}. \tag{6}$$

- *Robustness of state-to-state transfer.* For an initial state along $\hat{\mathbf{u}}_0$ on the Bloch sphere, the corresponding conditions are

$$\sum_{j=1}^{N} \epsilon_j \hat{\mathbf{m}}_j \parallel \hat{\mathbf{u}}_0 \qquad \text{(first order)}, \tag{7}$$

$$\sum_{k=2}^{N}\sum_{j=1}^{k-1} \epsilon_k \epsilon_j \hat{\mathbf{m}}_k \times \hat{\mathbf{m}}_j \parallel \hat{\mathbf{u}}_0 \qquad \text{(second order)}. \tag{8}$$